

MAFF-Net: Enhancing 3D Object Detection With 4D Radar via Multi-Assist Feature Fusion

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Abstract—Perception systems are crucial for the safe operation of autonomous vehicles, particularly for 3D object detection. While LiDAR-based methods are limited by adverse weather conditions, 4D radars offer promising all-weather capabilities. However, 4D radars introduce challenges such as extreme sparsity, noise, and limited geometric information in point clouds. To address these issues, we propose MAFF-Net, a novel multi-assist feature fusion network specifically designed for 3D object detection using a single 4D radar. We introduce a sparsity pillar attention (SPA) module to mitigate the effects of sparsity while ensuring a sufficient receptive field. Additionally, we design the cluster query cross-attention (CQCA) module, which uses velocity-based clustered features as queries in the cross-attention fusion process. This helps the network enrich feature representations of potential objects while reducing measurement errors caused by angular resolution and multipath effects. Furthermore, we develop a cylindrical denoising assist (CDA) module to reduce noise interference, improving the accuracy of 3D bounding box predictions. Experiments on the VoD and TJ4DRadSet datasets demonstrate that MAFF-Net achieves state-of-the-art performance, outperforming 16-layer LiDAR systems and operating at over 17.9 FPS, making it suitable for real-time detection in autonomous vehicles.

Index Terms—Object detection, autonomous vehicle navigation, 4D radar, point cloud, deep learning methods.

I. INTRODUCTION

CURRENTLY, autonomous vehicles have emerged as a transformative mode of transportation for enhancing driving safety and efficiency [1]. To successfully avoid obstacles and prevent traffic accidents, 3D object detection, a core module of autonomous perception systems, has seen rapid advancements due to its ability to provide comprehensive scene understanding [2]. However, traffic accidents involving autonomous

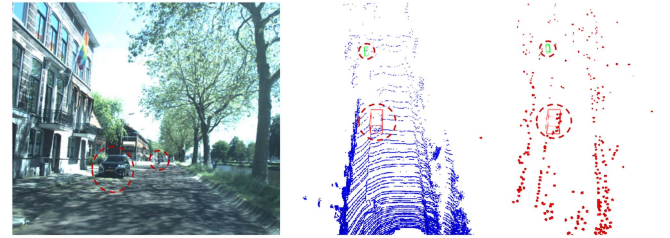


Fig. 1. Comparison of 4D radar and LiDAR point clouds from the View-of-Delft (VoD) dataset [5]. The first column shows scene images, the second shows LiDAR points, and the third presents single-scan 4D radar points.

vehicles continue to rise under extreme weather conditions, highlighting persistent challenges to the effectiveness of 3D object detection [3]. Optical sensors such as LiDAR and cameras are particularly susceptible to adverse weather, resulting in significant degradation in detection performance. Against this backdrop, radar, which operates using electromagnetic waves, is gaining increasing attention in autonomous vehicle perception systems due to its cost-effectiveness, all-weather operability, and precise velocity measurement capabilities. In particular, with the advent of high-resolution 4D radar sensors, radar has become a promising solution for achieving reliable 3D object detection under challenging weather conditions.

4D radar data formats can be broadly categorized into 4D tensors [4] and point clouds [5], [6]. Due to the compact representation and efficient processing of point cloud formats, they have become the preferred choice for real-time 3D object detection in autonomous vehicles. Additionally, the similarity between 4D radar point cloud data and LiDAR enables the direct application of deep learning methods originally developed for LiDAR [7], [8]. However, as shown in Fig. 1, 4D radar point clouds are significantly sparser than LiDAR point clouds, with the number of points being only about 1% of LiDAR in the same field of view, and they suffer from considerable noise, primarily due to speckle noise and multipath reflections [9]. Therefore, directly applying LiDAR-based 3D object detection methods to 4D radar data often yields suboptimal results. While recent methods, such as SMURF [10], RadarPillars [11], and MUFASA [12], have proposed promising approaches to narrow this gap, their detection performance remains unsatisfactory.

To address these challenges, we conducted an in-depth study of the characteristics of 4D radar point cloud data and proposed a multi-assist feature fusion network (MAFF-Net), a 3D object detection network tailored explicitly for 4D radar data. MAFF-Net aims to efficiently and accurately detect 3D objects

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Source code is available at <https://github.com/TRV-Lab/MAFF-Net>.

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in sparse and noisy 4D radar point clouds. Our contributions can be summarized as follows:

- We propose a sparsity pillar attention (SPA) module that extracts global features by interacting sparse pillars with embedded features, ensuring a sufficient receptive field while maintaining efficiency and real-time performance.
- We designed a cluster query cross-attention (CQCA) module that utilizes the feature map, formed by clustering spatial and velocity information, as queries in the cross-attention fusion. This improves object detection and mitigates noise effects.
- We introduce a cylindrical denoising assist (CDA) module, which effectively aggregates features through cylindrical constraints combined with the proposal's positional information, further reducing noise interference.
- Experimental results demonstrate that our MAFF-Net outperforms existing state-of-the-art (SOTA) 4D radar 3D object detection methods, exhibiting strong generalization capabilities and meeting the real-time detection requirements of autonomous vehicles.

II. RELATED WORKS

A. 3D Object Detection With Cameras

In recent research, camera-based 3D object detection has been extensively studied. In particular, the introduction of the “Lift, Splat, Shoot” (LSS) [13] framework, which unifies multi-view information into bird’s-eye view (BEV), has significantly enhanced the performance of camera-based 3D object detection. Furthermore, inspired by DETR [14] and Deformable DETR [15], transformer-based methods have gained prominence [16]. These approaches frequently employ Swin Transformer [17] as the backbone for feature extraction, leveraging the attention [18] mechanism to enable effective global feature interaction, which improves detection performance. Specifically, the recent introduction of RoPE-2D [19] has further enhanced the modeling of positional information during feature extraction. Building on these insights, we apply these techniques to the BEV feature maps of point clouds to achieve effective global feature interaction and expand the receptive field, thereby improving detection accuracy. However, due to the ambiguity in depth estimation, the performance of 3D object detection from camera images remains limited [20].

B. 3D Object Detection With LiDAR

Due to its accuracy in depth estimation, LiDAR-based 3D object detection methods have been widely applied in autonomous vehicle perception tasks. These methods can be categorized into three main approaches: point-based [21], voxel-based [8], [22], and point-voxel hybrid methods [23]. Voxel-based methods have gained widespread industrial application due to their excellent real-time performance. However, point-voxel hybrid methods offer an optimal solution to strike a better balance between accuracy and efficiency. While these approaches have significantly improved the performance of LiDAR-based 3D object detection, their limitations in adverse weather conditions remain, raising concerns about the overall safety of autonomous vehicle systems.

C. 3D Object Detection With 4D Radar

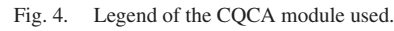
Radar has become an essential sensor in autonomous vehicles, valued for its low cost, long-range sensing, Doppler velocity measurement, and robustness in adverse weather, supporting tasks like segmentation [24], [25], detection, and especially 3D object detection. 4D radars provide denser point clouds and height information, elevating their role beyond merely serving as an auxiliary modality in multi-modal 3D object detection frameworks.

Autonomous vehicle datasets such as TJ4DRadSet [6] and VoD [5] have significantly driven advances in 4D radar-based 3D object detection. These datasets have set benchmarks using 3D object detection networks initially designed for LiDAR, such as SECOND [7] and PointPillars [8], demonstrating the effectiveness of voxel-based methods on the 4D radar. However, 4D radar point clouds are sparser and noisier than LiDAR, making accurate object geometry capture particularly challenging and resulting in lower detection performance. To bridge this gap, researchers are exploring various methods to enhance 4D radar features. For instance, SMURF [10] integrated a kernel density estimation branch to mitigate sparsity and noise by reducing measurement errors. RadarPillars [11] enhances detection performance by leveraging velocity features and introducing PillarAttention for sparse data processing. MUFASA [12] leverages GeoSPA and DEMVA to capture complex local and global information within each radar point cloud frame, enhancing feature extraction performance.

Moreover, integrating 4D radar with other sensors, such as cameras, is essential for 3D object detection. Whether through BEV fusion [20], [26] or multi-stage deep fusion [27], the performance of the 4D radar branch directly impacts the effectiveness of the fusion system. In particular, in adverse weather conditions where other sensors may fail, the performance of 4D radar is critical for maintaining system robustness. Further efforts are required to analyze and extract meaningful features from 4D radar data to enable more accurate 3D bounding box predictions. This study aims to develop a novel method that optimizes feature extraction from sparse 4D radar point clouds while minimizing noise.

III. THE PROPOSED METHOD

The proposed MAFF-Net architecture, shown in Fig. 2, predicts 3D bounding boxes from a single 4D radar point cloud. MAFF-Net consists of three components: the main branch, the assisted branch, and the detection head. In the main branch, we apply sparse pillar attention (SPA) to the BEV features generated from the raw point cloud using a pillar-based method [8], ensuring global interaction and a sufficient receptive field. The assisted branch introduces clustering query cross-attention (CQCA), using clustering feature assistance (CFA) to generate BEV queries for cross-attention fusion (CAF), which helps reduce noise and identify potential objects. We also design cylindrical denoising assistance (CDA), a sampling strategy inspired by cylindrical constraints, to filter noise and background points using the proposal’s positional information. Finally, fused BEV features are aggregated with clustered point cloud features at the keypoints’ locations, and a multi-task detection head predicts the 3D bounding boxes.



The sparse nature of the 2D pseudo-images generated by pillar feature encoding [8], known as pillar BEV features, results in only a limited number of effective features. While multi-level downsampling can improve global information integration for object detection [28], it risks losing small object details. To address this, we propose a sparsity pillar attention (SPA) module, shown in Fig. 3, which leverages the local sparsity of pillar features and incorporates rotational positional encoding (RoPE) [19] to enhance spatial perception.

where C_h represents half of the number of channels C , and i denotes the channel index, with a range of $[0, C_h - 1]$.

as follows:

$$\begin{bmatrix} F'_1 \\ F'_2 \end{bmatrix} = \begin{bmatrix} \cos(h_j \theta_i) \cos(w_j \theta_i) & \sin(h_j \theta_i) \sin(w_j \theta_i) \\ -\sin(h_j \theta_i) \sin(w_j \theta_i) & \cos(h_j \theta_i) \cos(w_j \theta_i) \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \end{bmatrix}. \quad (2)$$

To enhance computational efficiency, we utilize a sparse mask to capture non-empty features. These non-empty features are obtained by applying a sparse mask to the flattened result of adding the RoPE to the pillar BEV feature $F_{BEV}^{rot} = [F_1^T \ F_2^T]^T$. This approach reduces the computational complexity of extracting BEV feature maps, lowering it from HW to M .

Subsequently, we apply standard self-attention [18] to learn keys (K), queries (Q), and values (V), as the self-attention mechanism effectively captures long-range dependencies to enrich feature representation. Finally, the sparse mask indexes and reshaping are used to recover the pillar BEV feature $F_{BEV}^P \in \mathbb{R}^{H \times W \times C}$.

To address the noise and outliers in 4D radar point clouds, we designed the clustering query cross-attention (CQCA) module, shown in Fig. 4, to filter disruptive points and help the network learn effective object locations. The module includes clustering feature assist (CFA) and cross-attention fusion (CAF).

In CFA, we first extract three key features $\mathbf{X} \in \mathbb{R}^3$ from the raw point clouds for object detection: (x, y, v_r) , where v_r represents the absolute radial velocity. These key features are chosen because they effectively cluster the object and remove

noise points. Compared to the instability of RCS, v_r is more useful for clustering. Since radar point clouds are often clustered around objects, we designed a growth-based clustering algorithm (GCA), similar to DBSCAN [29], which clusters the point cloud using BEV plane coordinates (x, y) . The key difference is that GCA filters the clustered points using v_r , as points within the same cluster generally have consistent velocities. In DBSCAN, Eps defines the search radius, and $MinPts$ specifies the minimum points to form a cluster. We determine these parameters based on the angular resolution and point cloud density of the 4D radar. Additionally, we suppress large clusters to avoid redundant point clouds from static objects like buildings or fences.

Next, the final output of our GCA combines the number of points in each cluster, label, and three key features, resulting in new points $\mathbf{X}' \in \mathbb{R}^5$, where the five dimensions correspond to the cluster size, label, and the (x, y, v_r) features. This representation is then projected onto a BEV map of size $H \times W$. After passing through multiple convolutional layers (Conv2D), batch normalization (BN), and ReLU activation functions (CBR), we generate the clustered BEV feature F_{BEV}^C , which shares the same tensor size as F_{BEV}^P . The CFA process can be expressed with the following equation:

$$\begin{aligned} F_{BEV}^C &= \text{CBR}(\text{Project}(\mathbf{X}')), \\ \mathbf{X}' &= \text{Cat}(\text{DBSCAN}(\mathbf{X}), \mathbf{X}), \end{aligned} \quad (3)$$

where Cat denotes concatenation.

In CAF, inspired by ICAFusion [30], we fuse the two BEV feature maps, F_{BEV}^P and F_{BEV}^C , through the CAF mechanism. The clustered BEV feature F_{BEV}^C removes noise points and highlights potential regions containing objects. The flattened token T_C serves as the Q, while the flattened token T_P is used as the K and V in the attention mechanism, guiding the model to focus on specific feature areas. We apply a standard attention mechanism and learn the trainable weights $\delta_1, \delta_2, \delta_3$, and δ_4 , initializing these weights to 1 during training. Subsequently, the token T_{PC} is reshaped into the BEV feature map, and we use a squeeze-and-excitation layer (SE) [31] to help the network concentrate on specific fused features. Next, we concatenate the feature map processed by the SE with the clustered BEV feature and apply CBR to obtain the fused BEV feature F_{BEV}^{PC} . The CAF process is expressed by the following equation:

$$\begin{aligned} F_{BEV}^{PC} &= \text{CBR}(\text{Cat}(\text{SE}(\text{Reshape}(T_{PC}) + F_{BEV}^P), F_{BEV}^C)), \\ T_{PC} &= \delta_3 \cdot Z'_T + \delta_4 \cdot \text{FFN}(Z'_T), \\ Z'_T &= \delta_1 \cdot T_P + \delta_2 \cdot Z_T W^O, \\ Z_T &= \text{softmax}\left(\frac{Q_C K_P^T}{\sqrt{d_k}}\right) V_P, \end{aligned} \quad (4)$$

where Z'_T and Z_T are intermediate computation vectors.

Finally, after processing the fused BEV feature, we apply a 2D backbone network and a region proposal network (RPN) to generate 3D proposals. Specifically, similar to the RPN used in PointPillars [32], we utilize a multi-scale feature fusion module and a 2D detection head to produce high-quality 3D proposals.

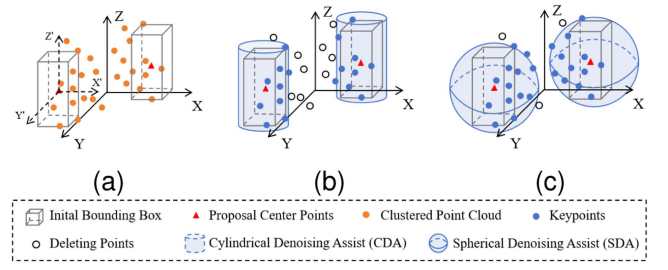


Fig. 5. Illustration of keypoints sampling for different denoising strategies: (a) keypoints before denoising; (b) keypoints after CDA; (c) keypoints after SDA.

C. Cylindrical Denoising Assist Module

Considering the diversity of traffic scenes, we sample the entire scene into keypoints to refine the fused BEV features. However, due to speckle and multipath reflections, 4D radar detects many noise points that hold little practical significance for feature representation and object detection. Therefore, we propose a cylindrical denoising assist (CDA) module that imposes cylindrical constraints on the sampled space to generate keypoints around 3D objects, thereby effectively enhancing the quality of feature representation.

In the CDA module, we input the 3D coordinates $\mathcal{P} = \{p_1, p_2, \dots, p_n\} \in \mathbb{R}^3$ of the clustered point cloud and the 3D proposals $B = \{O_i, l_i, w_i, h_i\}$, where $O_i \in \mathbb{R}^3$ denotes the 3D center coordinates, and l_i, w_i , and h_i represent the length, width, and height of the 3D proposals, respectively. Next, as shown in Fig. 5(a), we transform the coordinates to the center of each 3D proposal and fit the space of the 3D object using a cylinder CY , as follows:

$$\begin{aligned} CY &= \{O_i, r_i, e_i\}, \\ r_i &= \max(l_i, w_i) \times 0.5 + r^{(s)}, \\ e_i &= h_i + r^{(s)}, \end{aligned} \quad (5)$$

where r_i and e_i represent the radius and height of the cylinder, while $r^{(s)}$ denotes the hyperparameter for the maximum expansion scale of the proposal. Since the cylindrical 3D spatial region encapsulates the 3D proposal, we can directly select all points inside the cylinder as keypoints $\mathcal{P}'$ from the clustered point cloud $\mathcal{P}$ using a straightforward conditional formula as follows:

$$\mathcal{P}' = \{p_i | x_i^2 + y_i^2 \leq r_j^2, z_i \in [-e_j/2, e_j/2]\}, \quad (6)$$

where $[r_j, e_j] \in CY$, $[x_i, y_i, z_i] \in n_i$. As shown in Fig. 5(b), we obtain a certain number of denoised keypoints S_1 from the clustered point cloud using the cylinder. However, to achieve a uniform sampling that meets a fixed total of S keypoints, we sample the remaining $(S - S_1)$ keypoints from the raw points. The remaining keypoints are sampled using the farthest point sampling algorithm. In Fig. 5(c), we show the spherical denoising assist (SDA), which has a larger range but may include irrelevant points, potentially hindering effective denoising. This is confirmed by the ablation study, highlighting the superiority of our method.

By excluding noise points, the CDA strategy obtains more keypoints belonging to 3D objects. Similar to [23], through

point-pillar set abstraction, we aggregate features from the clustered point cloud and the fused BEV feature at the key-points' locations, and then concatenate these features along the channel dimension to generate pillar-point features. This feature combines the advantages of both voxel-based and point-based methods, enabling high-quality 3D object detection. Finally, we input the RPN-based 3D proposals along with the pillar-point features obtained through the RoI-grid pooling operation into the detection head.

D. Detection Head

After the CDA module, we employ the ROIHead [23] to perform precise 3D bounding box prediction. These predictions include essential elements such as the orientation angle, three-dimensional size (width, length, and height), center point, and confidence score. MAFF-Net aims to generate classification scores based on these predicted 3D bounding boxes. The detection head consists of two primary components: the classification loss L_{cls} [33] and the regression loss L_{reg} [34]. To balance the distribution of positive and negative samples within the network, we apply a variant of focal loss in the loss calculation for the detector. The regression loss L_{reg} is computed using the smooth $L1$ loss. Finally, the overall loss function for our MAFF-Net can be defined as:

$$L_{total} = L_{cls} + L_{reg}. \quad (7)$$

IV. EXPERIMENTS AND ANALYSIS

A. Datasets and Evaluation Metrics

We evaluated the performance of our method using the popular VoD [5] and TJ4DRadSet [6] datasets, both of which provide 4D radar data in point cloud format.

The VoD [5] dataset offers three types of 4D radar point clouds: single-scan, three-scan, and five-scan. We used the five-scan data to detect three object categories: cars, pedestrians, and cyclists. Following the VoD evaluation script, we analyzed the entire annotated area and the driving corridor. The entire annotated area covers all labeled objects within the camera FoV up to 50 meters, while the driving corridor focuses only on objects relevant to the vehicle's driving path. Metrics include 3D average precision (AP_{3D}) for each object category, mean 3D AP (mAP_{3D}), and mean BEV AP (mAP_{BEV}). The intersection over union (IoU) thresholds were 0.5 for cars and 0.25 for pedestrians and cyclists.

The TJ4DRadSet [6] dataset provides 3D bounding box annotations for objects at greater distances and in open highway scenes. Its evaluation metrics are similar to those of the VoD dataset but allow specifying evaluation regions based on sensor distance. We focus on objects within 70 meters of the radar, using IoU thresholds of 0.5 for cars and trucks, and 0.25 for pedestrians and cyclists.

B. Experimental Setup

We built our MAFF-Net model using OpenPCDet [37], an open-source 3D detection toolbox based on PyTorch.

Hyper-parameter Settings: For the VoD dataset, the 4D radar point cloud has a range of X [0, 51.2], Y [−25.6, 25.6], and Z [−3, 2], while the TJ4DRadSet dataset spans X [0, 69.12], Y

[−39.68, 39.68], and Z [−4, 2], all in meters. In both datasets, each pillar covers 0.16 square meters, supporting up to 16 points per pillar.

In our MAFF-Net model, we set $C = 64$ and $E = 64$ for all modules. For the CDA, the maximum expansion radius $r^{(s)}$ is 2 m for the VoD dataset and 2.5 m for the TJ4DRadSet dataset. In the CQCA, Eps and $MinPts$ are adjusted based on radar performance. For the VoD dataset, Eps is 0.4 m and $MinPts$ is 10 points, while for the TJ4DRadSet dataset, Eps is 2 m and $MinPts$ is 4 points. The BEV feature map resolution $H \times W$ is determined by the sparsity characteristics of 4D radar point clouds to balance object size and accuracy. It is set to 320×320 for the VoD dataset and 496×432 for the TJ4DRadSet dataset.

Additionally, in the 3D proposals generated by the RPN within the CQCA, we randomly sampled 128 proposals with a 1:1 ratio for positive and negative proposals. In the CDA, the total number of sampled keypoints S is set to 1024.

Training Details: We implemented the model on an Ubuntu 22.04 server with an Intel Core i9-13900 K CPU, 64 GB RAM, and an NVIDIA RTX 4090 24 GB GPU, using PyTorch 1.10.1. The training spanned 60 epochs with a batch size of 4, using the Adam optimizer with a learning rate of 0.01 and cosine annealing learning rate decay. The inference speed was measured with a batch size of 1, a single NVIDIA RTX 4090 GPU, and FP16 precision.

We applied data augmentation, including random X-axis flipping (50% chance), global scaling (0.95–1.05), and random Z-axis rotation ($\pm 0.7854\pi$ radians).

C. Comparison With State-of-the-art

VoD Result: In Table I, we compare MAFF-Net with existing methods on the VoD validation set. MAFF-Net surpasses current SOTA methods, establishing a new benchmark. MAFF-Net improves mAP_{3D} and mAP_{BEV} across the entire annotated area by 3.62% and 1.60%, respectively. In the driving corridor, these metric improvements are 2.00% and 1.93%. Notably, MAFF-Net excels in detecting small objects, such as pedestrians, with an AP improvement of at least 4.64%. These results demonstrate MAFF-Net's ability to extract object features from sparse point clouds and enhance detection accuracy. Moreover, the results reveal that PV-RCNN(V) outperforms PV-RCNN(PP), indicating that the Pillar-based backbone is more effective than the Voxel-based backbone in extracting features from 4D radar point clouds. To further illustrate the breakthrough performance of MAFF-Net and its potential significance for autonomous driving, we compared it to LiDAR-based 3D object detection methods. As shown in Table III, We adopt PointPillars as the benchmark for LiDAR methods, as it forms the backbone of our approach. In the experiments, we downsampled the 64-layer LiDAR data from the VoD dataset to 32-layer, 16-layer, and 8-layer versions for evaluation. The results indicate that, even with fewer input points, MAFF-Net enables the 4D radar to outperform the 16-layer LiDAR in detection accuracy. Therefore, with advancements in 4D radar technology, it has the potential to become a primary sensor for 3D object detection in autonomous vehicles.

TJ4DRadSet Result: To further validate the effectiveness and generalization of MAFF-Net, we trained the model on the TJ4DRadSet dataset, with the results on the test set presented

TABLE I
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE VOD [5] VALIDATION SET

Method	AP in the Entire Annotated Area(%)					AP in the Driving Corridor(%)					FPS(Hz)
	Car	Pedestrian	Cyclist	mAP _{3D}	mAP _{BEV}	Car	Pedestrian	Cyclist	mAP _{3D}	mAP _{BEV}	
PointPillars [8]	36.93	35.72	63.31	45.32	51.81	70.87	46.87	84.99	67.58	71.17	187.0
CenterPoint [35]	32.74	38.00	65.51	45.42	52.75	62.01	48.18	84.98	65.06	67.05	72.2
PillarNeXt [22]	30.81	33.11	62.78	42.23	52.93	66.72	39.03	85.08	63.61	68.35	67.2
RadarPillarNet [26]	39.30	35.10	63.63	46.01	N/A	71.65	42.80	83.14	65.86	N/A	N/A
PV-RCNN(V) [23]	38.01	41.16	65.56	48.24	50.24	71.21	51.75	85.12	69.36	72.23	43.5
PV-RCNN(PP) [‡] [23]	40.11	42.11	67.58	49.93	55.93	72.14	52.56	86.29	70.33	73.34	52.4
SMURF [10]	42.31	39.09	71.50	50.97	56.77	71.74	50.54	86.87	69.72	72.05	30.3*
RadarPillars [11]	41.10	38.60	72.60	50.70	N/A	71.10	52.30	87.90	70.50	N/A	179.1*
MUFASA [12]	43.10	38.97	68.65	50.24	N/A	72.50	50.28	88.51	70.43	N/A	N/A
MAFF-Net(ours)	42.33	46.75	74.72	54.59	58.37	72.28	57.81	87.40	72.50	75.27	28.7
Improvement	-0.77	+4.64	+2.12	+3.62	+1.60	-0.22	+5.25	-1.11	+2.00	+1.93	-

PV-RCNN(V) and PV-RCNN(PP) use SECOND [7] and PointPillars [8] as backbones, respectively. The [‡] symbol indicates the baseline model, and * denotes inference speeds from referenced sources. The best values are in bold.

TABLE II
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE TJ4DRADSET [6] TEST SET

Method	AP _{3D} (%)					AP _{BEV} (%)					FPS(Hz)
	Car	Pedestrian	Cyclist	Truck	mAP	Car	Pedestrian	Cyclist	Truck	mAP	
PointPillars [8]	22.67	26.16	49.72	15.47	28.51	35.77	32.71	53.99	24.83	36.57	156.0
CenterPoint [35]	22.03	25.02	53.32	15.92	29.07	33.03	27.87	58.74	25.09	36.18	50.3
PillarNeXt [22]	22.33	23.48	53.01	17.99	29.20	36.84	25.17	57.07	23.76	35.71	48.5
RadarPillarNet [26]	28.45	26.24	51.57	15.20	30.37	45.72	29.19	56.89	25.17	39.24	N/A
PV-RCNN(V) [23]	25.31	20.00	47.98	15.08	27.09	31.47	20.25	53.45	19.79	31.24	29.5
PV-RCNN(PP) [‡] [23]	24.79	21.11	45.34	23.49	28.68	38.10	25.62	46.48	32.14	35.58	48.1
SMURF [10]	28.47	26.22	54.61	22.64	32.99	43.13	29.19	58.81	32.80	40.98	23.1*
MUFASA [12]	23.56	23.70	48.39	25.25	30.23	41.25	24.54	53.64	36.97	39.10	N/A
MAFF-Net(ours)	27.31	33.13	54.35	26.71	35.38	39.05	35.25	56.35	35.73	41.59	17.9
Improvement	-1.16	+6.89	-0.26	+1.46	+2.39	-6.67	+2.54	-2.46	-1.24	+0.61	-

PV-RCNN(V) and PV-RCNN(PP) use SECOND [7] and PointPillars [8] as backbones, respectively. The [‡] symbol indicates the baseline model, and * denotes inference speeds from referenced sources. The best values are in bold.

TABLE III
COMPARISON OF 4D RADAR AND LiDAR RESULTS ON THE VOD [5] VALIDATION SET

Method	M	Spe	Input Points	AP in the Entire Annotated Area(%)		AP in the Driving Corridor(%)	
				mAP _{3D}	mAP _{BEV}	mAP _{3D}	mAP _{BEV}
PointPillars [8]	L	64-layer	18000 ~ 30000	61.60	64.08	78.93	79.10
		32-layer	9000 ~ 15000	57.31	59.40	77.88	78.38
		16-layer	4500 ~ 7500	51.70	54.79	72.05	75.84
		8-layer	2250 ~ 3750	49.74	52.90	70.65	73.85
MAFF-Net(ours)	R	5-scan	100 ~ 2,800	54.59	58.37	72.50	75.27

M represents modality, L represents LiDAR, R represents 4D radar, and Spe represents specifications. Input points refer to the number of points input into the model from the LiDAR and 4D radar within the same FoV.

in Table II. The experimental results indicate that MAFF-Net achieves a 2.39% higher mAP_{3D} than the SOTA method. For mAP_{BEV}, MAFF-Net reaches 41.59%, establishing a new SOTA performance.

Finally, to assess the potential of MAFF-Net for real-time object detection in autonomous driving, we evaluated its inference speed on the VoD and TJ4DRadSet datasets. The results show that MAFF-Net achieves a minimum inference speed of 17.9 frames per second (FPS), surpassing the 4D radar data acquisition rate (~15 FPS). Moreover, as a frame rate of 10 FPS is generally sufficient for real-time applications [36], these findings confirm that MAFF-Net meets the requirements for real-time object detection.

D. Ablation Studies

To validate the effectiveness of each module in MAFF-Net, we performed ablation experiments on the VoD validation set and

the TJ4DRadSet test set, specifically analyzing the contributions of the SPA, CQCA, and CDA modules.

MAFF-Net builds upon PV-RCNN [23] by integrating several key modules described in Section II. Using PV-RCNN (PP) as the baseline model, Table IV shows that each module significantly enhances the network's performance across the entire annotated area and the driving corridor, especially in mAP_{3D} and mAP_{BEV} metrics across all categories. Notably, the SPA and CQCA modules remarkably improve the detection of small objects like pedestrians and cyclists. These results demonstrate the ability of the SPA and CQCA modules to effectively suppress noise and extract critical features from sparse point clouds. Replacing the farthest point sampling module in PV-RCNN with the proposed CDA module yields additional performance gains, demonstrating CDA's superior ability to suppress noise and select keypoints from 4D radar point clouds, thereby enhancing the accuracy of 3D bounding box predictions. Furthermore, analyzing different combinations of the three modules reveals synergy, leading to balanced and improved overall performance.

To further validate the effectiveness of the SPA and CQCA modules, we conducted ablation experiments using PointPillars [8] as the baseline. Table V demonstrates that both modules improve detection performance, consistent with the results of the PV-RCNN(PP) experiments.

To evaluate the advantages of cylindrical keypoint sampling over spherical keypoint sampling for 4D radar data, we compared the CDA and SDA methods. As shown in Table VI, CDA outperforms SDA in overall performance.

TABLE IV
ABLATION STUDY ON THE VoD [5] VALIDATION AND TJ4DRadSet [6] TEST SETS, USING PV-RCNN(PP) [23] AS BASELINE

Module			AP in the Entire Annotated Area(%)					AP in the Driving Corridor(%)				
SPA	CQCA	CDA	Car	Pedestrian	Cyclist	mAP _{3D}	mAP _{BEV}	Car	Pedestrian	Cyclist	mAP _{3D}	mAP _{BEV}
✓			40.11	42.11	67.58	49.93	55.93	72.14	52.56	86.29	70.33	73.34
✓	✓		41.51	45.87	66.09	51.16	57.14	72.08	57.43	84.75	71.42	75.06
✓			40.29	45.96	74.46	53.59	57.47	71.87	57.27	87.15	72.10	75.07
✓		✓	41.91	45.32	69.13	52.12	56.94	71.45	56.62	85.91	71.33	73.95
✓	✓	✓	41.73	45.90	74.19	53.94	57.72	72.21	55.99	87.33	71.84	74.71
✓	✓	✓	42.33	46.75	74.72	54.59	58.37	72.28	57.81	87.40	72.50	75.27
<i>Baseline Improvement</i>			+2.22	+4.64	+7.14	+4.66	+2.44	+0.14	+5.25	+1.11	+2.17	+1.93

Module			AP _{3D} (%)					AP _{BEV} (%)				
SPA	CQCA	CDA	Car	Pedestrian	Cyclist	Truck	mAP	Car	Pedestrian	Cyclist	Truck	mAP
✓			24.79	21.11	45.34	23.49	28.68	38.10	25.62	46.48	32.14	35.58
✓	✓		26.27	29.63	48.05	24.04	32.00	38.07	33.23	53.85	28.81	38.49
✓			27.92	29.37	53.82	25.62	34.18	39.96	33.15	54.71	35.64	40.87
✓		✓	27.03	30.01	50.10	23.53	32.67	39.51	35.03	55.27	29.56	39.84
✓	✓	✓	29.76	26.99	54.11	28.45	34.83	39.91	32.16	55.42	36.33	40.96
✓	✓	✓	27.31	33.13	54.35	26.71	35.38	39.05	35.25	56.35	35.73	41.59
<i>Baseline Improvement</i>			+2.52	+12.02	+9.01	+3.22	+6.70	+0.95	+9.63	+9.87	+3.59	+6.01

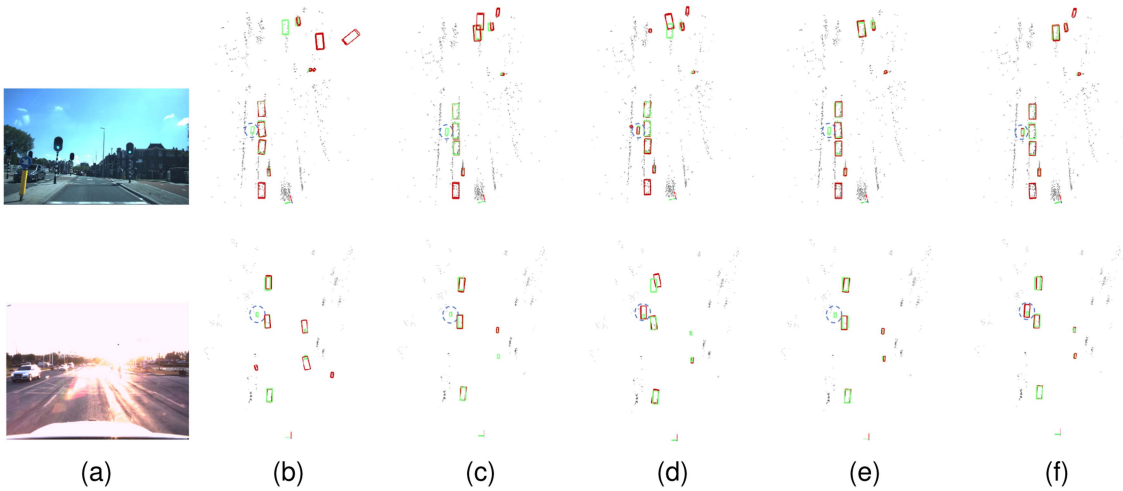


Fig. 6. Visualization results on the VoD validation set (first row) and TJ4DRadSet test set (second row). Each row shows a data frame with an image and radar points (gray points), and colored axes indicate the ego-vehicle's position. Green boxes represent ground truth, and red boxes represent predictions. (a) shows the image; (b) displays PointPillars results; (c) shows PV-RCNN(PP) baseline results; (d) shows results with the SPA module added to the baseline; (e) shows results with the CQCA module added to the baseline; (f) presents MAFF-Net (ours) results.

TABLE V
ABLATION STUDY ON THE VoD [5] VALIDATION SET WITH POINTPILLARS [8] AS THE BASELINE

Module		AP in the Entire Annotated Area(%)				AP in the Driving Corridor(%)			
SPA	CQCA	Car	Ped.	Cyc.	mAP _{3D}	Car	Ped.	Cyc.	mAP _{3D}
✓		36.93	35.72	63.31	45.32	70.87	46.87	84.99	67.58
✓		40.90	45.42	66.14	50.82	72.07	57.25	84.94	71.42
✓	✓	43.05	43.70	69.52	52.09	72.38	54.44	88.93	71.92
<i>Bas. Imp.</i>		+6.12	+7.98	+6.21	+6.77	+1.51	+7.57	+3.94	+4.34

TABLE VI
DENOISING STRATEGY EXPERIMENTS ON THE VoD [5] VALIDATION SET, WITH PV-RCNN (PP) [23] AS THE BASELINE

Strategy	AP in the Entire Annotated Area(%)				AP in the Driving Corridor(%)			
	Car	Ped.	Cyc.	mAP _{3D}	Car	Ped.	Cyc.	mAP _{3D}
SDA	40.11	42.11	67.58	49.93	72.14	52.56	86.29	70.33
CDA	41.31	42.90	66.15	50.12	72.05	55.48	83.73	70.42
	41.52	42.64	68.94	51.03	72.25	54.06	86.96	71.09

Fig. 6 presents sample visualization results of MAFF-Net on the VoD and TJ4DRadSet datasets. The qualitative results show

that MAFF-Net surpasses PointPillars and PV-RCNN(PP) in accurately predicting 3D bounding boxes for objects across diverse scenarios. Specifically, the SPA module enhances the detection of occluded objects through global feature interaction, while the CQCA module improves object localization and reduces false positives.

Overall, the results confirm the effectiveness of each proposed module in MAFF-Net and highlight the importance of noise suppression, global feature extraction, and keypoint selection for 3D object detection with 4D radar point clouds.

V. CONCLUSION

In this work, we propose MAFF-Net, a novel 4D radar-based 3D object detector designed to tackle the challenges posed by sparse, noisy, and semantically limited radar point clouds. MAFF-Net effectively suppresses noise and extracts meaningful features by leveraging spatial and geometric constraints. Experimental results on the VoD and TJ4DRadSet datasets demonstrate that MAFF-Net achieves SOTA performance, outperforming

16-layer LiDAR-based methods while maintaining real-time inference speed. Future research will integrate camera image data to enhance radar features, enabling more accurate and efficient 3D object detection.

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